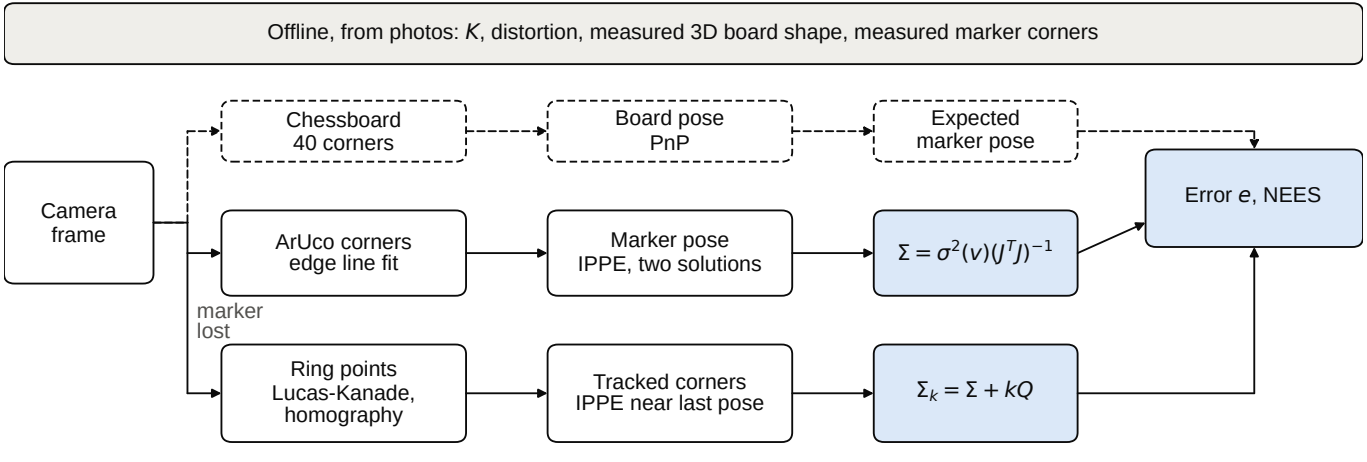
Offline, from photos: K, distortion, measured 3D board shape, measured marker corners
Chessboard
40 corners
Board pose
PnP
Expected
marker pose
Camera
frame
ArUco corners
edge line fit
Marker pose
IPPE, two solutions
$\Sigma = \sigma^2(v)(J^TJ)^{-1}$
Error e, NEES
marker
lost
Ring points
Lucas-Kanade,
homography
Tracked corners
IPPE near last pose
$\Sigma_k = \Sigma + kQ$
Dashed: reference, used only for evaluation